Rough basin boundaries are computationally hidden: the cross-boundary exponent of a latent-reasoning model

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Abstract

A basin boundary is *rough* if the cross-boundary Lyapunov exponent λ_x of the chaotic set it contains is smaller than the maximal exponent λ_1, and *filamentary* otherwise [2]. Both give the same uncertainty exponent κ/λ_1, so fractal dimensions cannot decide which case holds. We tried to decide it for the EqR Sudoku model, whose settle-time basins are fractal [1] with uncertainty exponent κ/λ_1 [7], and found a general obstruction instead. (i) Every observable we could build from generic perturbations is degenerate between the two cases, and statistics of survival or escape are confounded; five classes, each calibrated on toy systems of known type. (ii) Only probes along the exact cross direction separate them, at a precision cost we measure: the noise floor scales as $\varepsilon_{\text{mach}}^{h}$ and as $c^{h/(1-h)}$ for a direction error c, with $h = \lambda_x/\lambda_1$, over 37 decades of machine precision. (iii) With any feedback from the decision into the fast dynamics, the direction must also come from a genuine, kick-free orbit on the chaotic set; with such orbits the probe recovers λ_x (in toys). The resulting cost law puts EqR's answer at a genuine orbit of ∼450–850 loops in ∼460–860-digit arithmetic. Along the way we measure EqR's chaotic set directly: 7–11 positive exponents, an escape that leaks only 0.6–3% of the stretching, and escapes that happen as an abrupt "snap".

1 The question

For an open chaotic system with escape rate κ, the fraction of ε-close initial conditions with different outcomes scales as ε^{α} with $\alpha = \kappa/\lambda_{\max}$ [3,4,9]. For EqR we measured $\lambda_1 \approx 2.07$ per loop for the whole model, κ from 0.05 to 1.08 per loop depending on the puzzle, and the law on 9 puzzles (rank correlation 0.92) [7]. Bódai and Lucarini [2] showed that this law holds both for filamentary boundaries (a Cantor set of smooth sheets, $\lambda_x = \lambda_1$) and for rough ones (a nowhere-differentiable surface, $\lambda_x < \lambda_1$), and that fractal dimensions cannot tell them apart. One has to measure λ_x. Lai et al. [1] map fractal basins in reasoning models but do not measure λ_x (nor κ or λ_1).

2 EqR's chaotic set

Methods: true float64 (both precision fixes of [7]); long trajectories on the chaotic set by stagger-and-step [5] with backtracking; Lyapunov spectra by Benettin twins with QR; exact Jacobian–vector products by forward-mode differentiation (EqR's inner forward detaches its latents, so we call its recursion directly).

- **Spectrum.** 7–11 positive exponents on four puzzles (κ = 0.054, 0.118, 0.142, 0.246); e.g. puzzle 206: λ_k = 2.23, 1.68, 1.26, 1.01, 0.66, …, 0.05. B&L's necessary condition for roughness ($\lambda_2 > 0$) holds.
- **Leak.** $\sum \lambda^{+}$ = 6.4–8.5 per loop against κ = 0.05–0.25: by Kantz–Grassberger, escape costs 0.6–3% of the stretching.
- **Snap.** On the chaotic set ∼10 of 81 cells flip per loop. Escape is one loop in which 27–30 cells change and the latent jumps by ∼83%, followed by smooth convergence.
- **Where the churn lives.** No token group is an invariant subspace (every group leaks ≥53% of a perturbation after one loop), but the contested cells are partly self-contained (leakage 0.53–0.61 against 0.73–0.80 for random groups of the same size).
- **Puzzle dependence.** The spectra of puzzles 206, 225 and 80 agree within noise; 814 is reproducibly lower in λ_7–λ_{10} (two replicates).

3 Degeneracy beyond dimensions

Consider a skew structure: a fast "churn" with exponent λ_1 drives a decision coordinate with λ_x; the boundary is a graph over the churn with Hölder exponent $h = \lambda_x/\lambda_1$. A perturbation of size δ with a generic churn component lies $\sim \delta^h$ from the boundary, so the outcome changes after $\ln(1/\delta^h)/\lambda_x = \ln(1/\delta)/\lambda_1$ loops, exactly as for a filamentary boundary. Churn noise of size η reaches the decision as η^h, and the same cancellation applies. Hence every observable built from outcomes under generic perturbations has filamentary scaling. Table 1 lists the classes we tested on toy families of known type (rough, filamentary, dense spectra, hidden bases, fluctuating stretching).

4 Two precision laws

Along the exact cross direction the lifetime grows by $1/\lambda_x$ per e-fold of distance, against $1/\lambda_1$ along any generic line. This signal drowns below a floor. With rounding $\varepsilon_{\text{mach}}$, noise enters the churn and reaches the decision Hölder-amplified, giving $\delta_{\text{floor}} \sim \varepsilon_{\text{mach}}^{h}$. With a probe direction that is off by a fraction c toward the churn, $\delta_{\text{floor}} \sim c^{h/(1-h)}$. Fig. 1 tests both on a rotated

class	outcome on calibrated toys
bisection along lines	$1/\lambda_1$ in both cases (0.87 vs 0.92)
spectrum tests ("an exponent tracks κ")	work on toys; on EqR noise-limited and puzzle-dependent
survival vs. stretching	geometric confounds (sign flips)
topology of 2D slices	both look like stripes at fine scales
escape-triggered averages	relaxation confounds; a filamentary toy faked a rough rate $\approx \kappa$
exact cross direction	separates the cases (Sec. 4)

Table 1: Observables tested for distinguishing rough from filamentary boundaries.

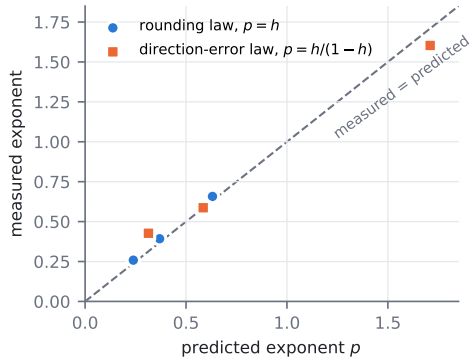


Figure 1: Measured scaling exponent of the noise floor against the prediction, for rounding (circles, three values of h) and for a direction error (squares).

Weierstrass-type toy ($x' = 3x \bmod 1$, $y' = ay + \cos 2\pi x$) in float16, float32, float64 and 25- and 40-digit arithmetic: 3/3 and 2/3 within 25% of the prediction (the miss falls on the favourable side). Both floors vanish as $h \to 0$: strongly rough boundaries are hidden at any fixed precision.

5 The orbit obstruction and the cost law

In an exactly skew toy the cross direction is the same at every point, and covariant Lyapunov vectors [6] computed along a kicked stagger trajectory recover λ_x (0.406 vs 0.405). EqR is not skew (Sec. 2). With feedback as weak as 10^{-3} from the decision into the churn, the same pipeline loses the signal and calls a rough toy filamentary (Fig. 2): the cross direction now depends on the orbit's own future, which a kicked pseudo-orbit does not follow. Genuine orbits of ∼160 steps, found by bisection [8] at 80 digits without any kick, restore it (9/9 verdicts correct, λ_x within 25% in 8/9).

Together: a one-decade probe window needs $c \le 0.01^{(1-h)/h}$, a genuine orbit of length $L \approx L_{\text{warm}} + \ln(1/c)/g$ on the chaotic set (g: spectral gap above λ_x), and $\sim L\lambda_1/\ln 10$ digits to hold it (toy check: 76 predicted, 80 used). For EqR puzzle 80 ($\lambda_1 = 2.34$; $h \approx 0.105$ if the boundary is skew-rough) and the measured gaps of 0.05–0.1, this means a genuine orbit of 450–850 loops in 460–860 digits: decidable in principle, out of reach in practice. In float64 no EqR puzzle is decidable: where stagger-and-step can hold an orbit

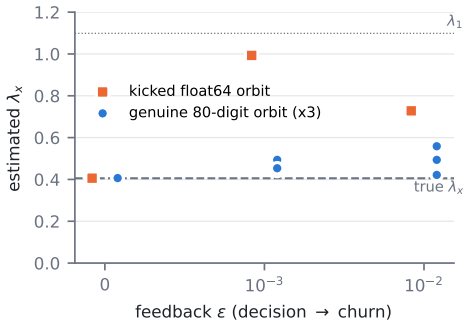


Figure 2: Estimated λ_x for a rough toy with feedback ϵ, from kicked float64 orbits (squares) and from genuine 80-digit orbits (circles, three each).

($\kappa \le 0.25$) the usable window is below a decade, and where it would open the orbit cannot be held.

6 Smaller models?

The cost does not grow with the size of the latent, so a small trained reasoner might be decidable. Three looped models trained on 4×4 Sudoku (a one-layer residual update with 32 or 64 dimensions; the same with several valid answers; a miniature EqR with 16 tokens and inner cycles, 94–95% accurate) never showed transient chaos: easy puzzles contract ($\lambda_1 < 0$), and the hardest either settle without an exponential tail or keep oscillating ($\lambda_1 \approx 0$). In this family, transient chaos appears to need scale.

7 Limitations

The degeneracy is an argument with numerical support, not a theorem. The laws are verified on toys of 2–30 dimensions. The EqR price assumes a skew-rough boundary ($h = \kappa/\lambda_1$). One model and one task. Scripts, pre-registrations (with every amendment and bug fix) and raw outputs exist for every number.

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